GeoAI Innovation Portfolio

Bridging Geographic Theory and Deep Learning Architectures on Apple Silicon

Foundational Vision

Computer Vision applied to earth observation and planetary geology.

Proj 1: LULC Classification

Visual Asset on Slide:

→ Grid of 13-band Sentinel-2 image patches.

→ Side-by-side comparison with CNN classification outputs.

Data Points to Display:

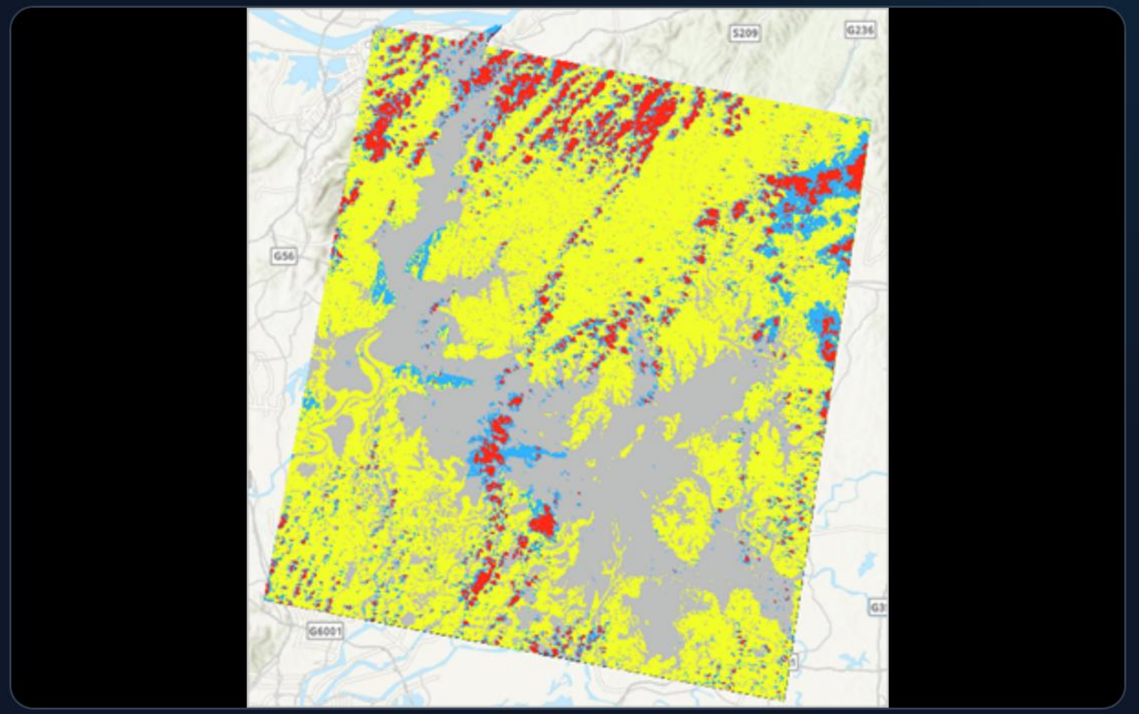
→ EuroSAT dataset volume & resolution.

→ MPS training speeds (Apple Silicon).

→ Top-1 Classification Accuracy metrics.

Presentation Bridge:

Highlight the use of NIR and SWIR bands to distinguish similar visual covers (e.g., turf vs. grass), automating manual remote sensing cartography.

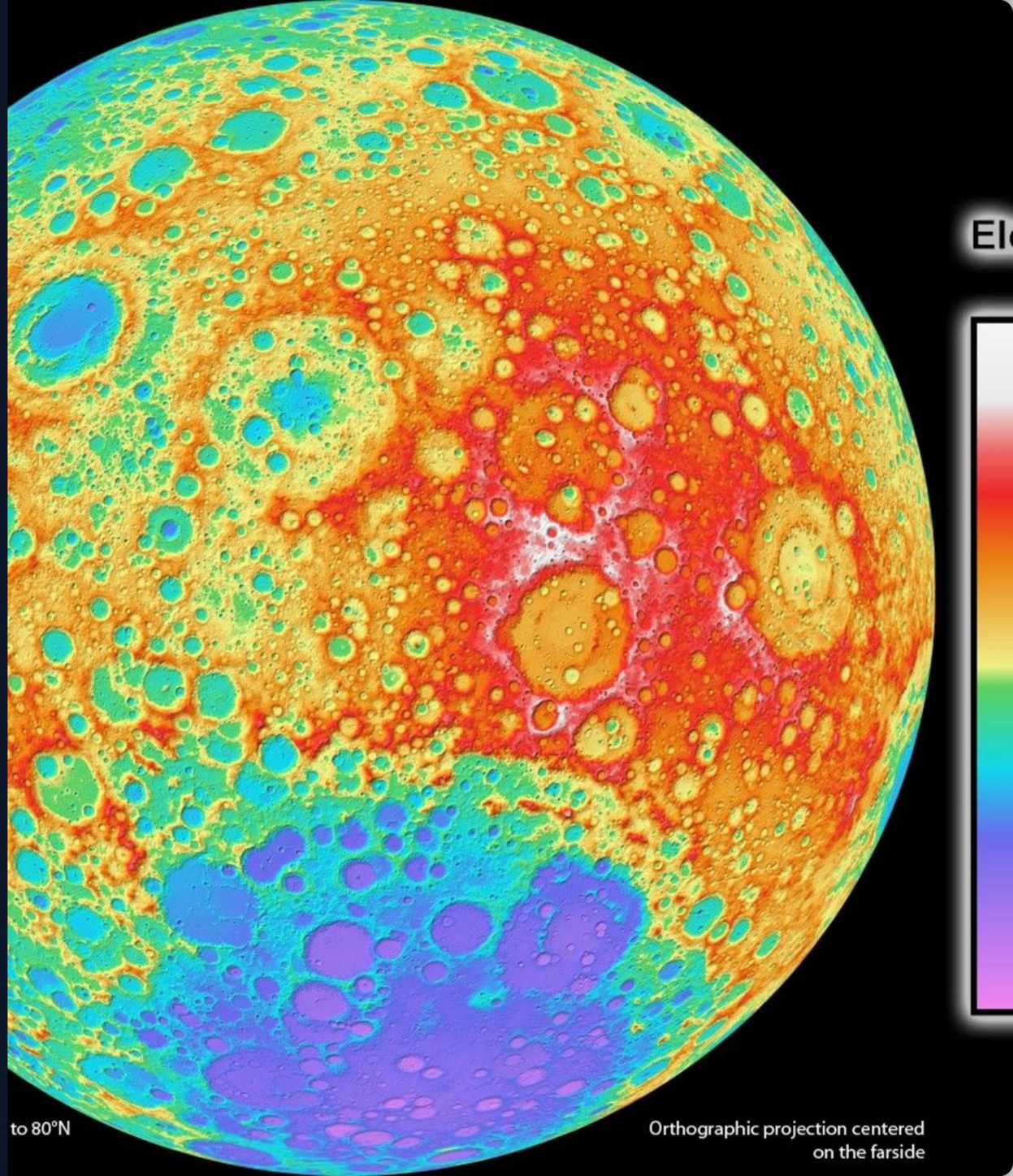


Proj 2: Crater Detection

Visual Asset: Heatmap of Lunar Reconnaissance Orbiter DEMs overlaying YOLOv8 bounding boxes.

Data Points: Precision/recall curves comparing Lunar testing vs. Mars zero-shot testing sets.

Core Novelty: Zero-shot inference domain adaptation from Lunar to Martian topography, bridging Geomorphology and Object Detection.



Advanced Spatial Analysis

Moving from basic classification to pixel-level mapping and temporal forecasting.

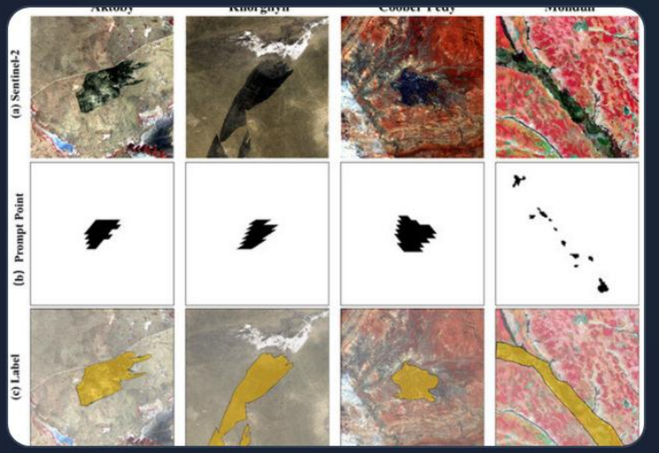
Proj 3: Wildfire Segmentation



Visual: Pre-fire canopy.



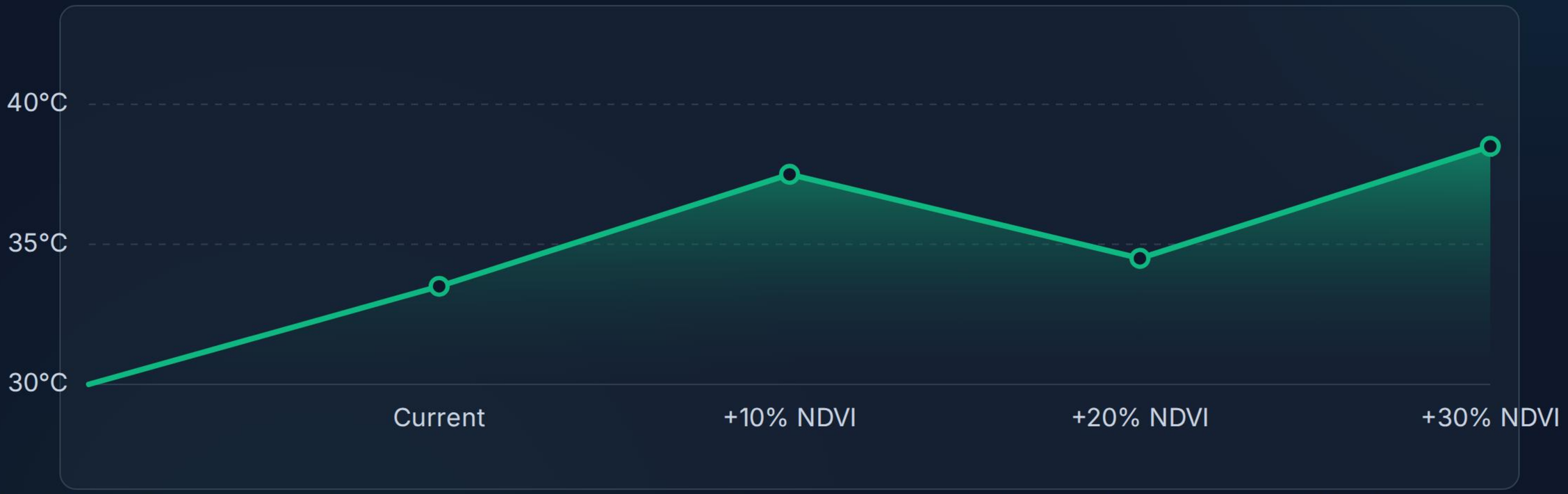
Visual: Post-fire destruction.



Visual: U-Net Semantic Mask.

Presentation Bridge: Explain the injection of the calculated dNBR delta directly into the U-Net input tensor. This physics-guided approach separates true ecological destruction from seasonal browning.

Proj 4: UHI Forecasting



Visual Asset: Chart showing forecasted surface temperature drops associated with simulated increases in neighborhood vegetation (NDVI) via LSTM manipulation.

Physics & Graph Networks

Incorporating thermodynamic laws and non-euclidean transport topologies.

Proj 5: Climate Downscaling

Physics-Informed SRGAN

Visual Asset: Side-by-side comparison of 30km ERA5 data vs. 1km SRGAN output generation.

Data Points: Root Mean Square Error (RMSE) and Peak Signal-to-Noise Ratio (PSNR) vs. baseline interpolation.

Thermodynamic Loss

Core Novelty: Instead of simple image upsampling, a custom physical loss function enforces mass conservation of precipitation.

$$L_{total} = L_{MSE} + \lambda L_{physics}$$

Proj 6: Human Mobility

Pre-Pandemic

Centralized commuting flows.

ST-GNN Processing

OpenStreetMap topologies processed as edges.

Post-Pandemic

Decentralized, suburban-heavy mobility patterns emerge.

Future Congestion

Predicting bottlenecks via edge density metrics.

Visual Asset: Network graph animation showing shifting edge weights on map topologies over time.

Ecological & Generative Frontiers

Predicting the future of the biosphere and engineering the urban environment.

Proj 7: Biodiversity



Biogeographical Shifts

- → **Visual Asset:** Forecast map highlighting predicted latitudinal migrations for 2070 under RCP 8.5 scenarios.
- → **Data Points:** WorldClim temperature deltas matched with GBIF occurrence probabilities.
- → **Core Novelty:** Multimodal ensemble modeling utilizing a CNN for habitat extraction and an MLP for climate data processing.

Proj 8: Synthetic Urban Plans

Visual Asset: High-res conditional diffusion outputs.

Showing original city → 2m sea level rise → 30% green space transformation.

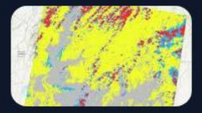
Data Points: DDPM inference speeds on M4, Fréchet Inception Distance (FID).

Core Novelty: Photorealistic counterfactual generation for policy stakeholders.

Questions?

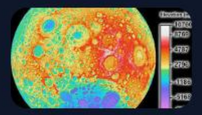
Thank you for exploring the GeoAI portfolio.

Image Sources



https://learn.arcgis.com/en/projects/classify-land-cover-to-measure-shrinking-lakes/GUID-8DD04D9F-BD34-4F1C-96E0-B3AAF69A2F22-web.png

Source: learn.arcgis.com



https://assets.science.nasa.gov/dynamicimage/assets/science/psd/lunar-science/2023/08/elevation_map.jpg?w=1277&h=1000&fit=clip&crop=faces%2Cfocalpoint

Source: science.nasa.gov



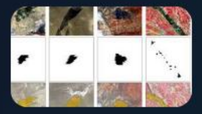
https://i.ytimg.com/vi/HIoy3mBBkDk/hq720.jpg?sqp=-oaymwEhCK4FEIIDSFryq4qpAxMIARUAAAAAGAEIAADIQj0AgKJD&rs=AOn4CLD2MtzI-IjVpS2I9fGdT73UVGQXEA

Source: www.youtube.com



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Image Sources



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